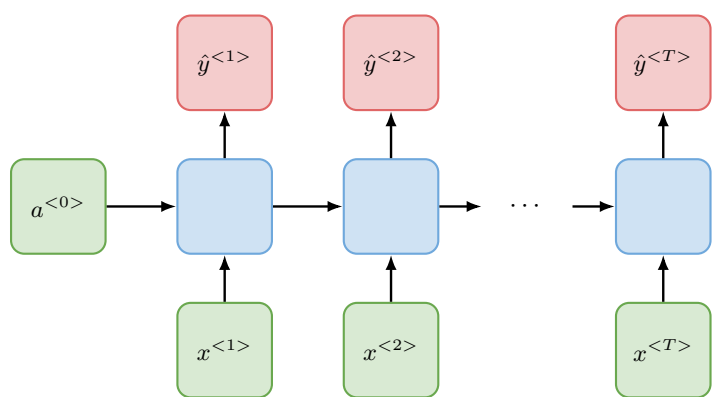
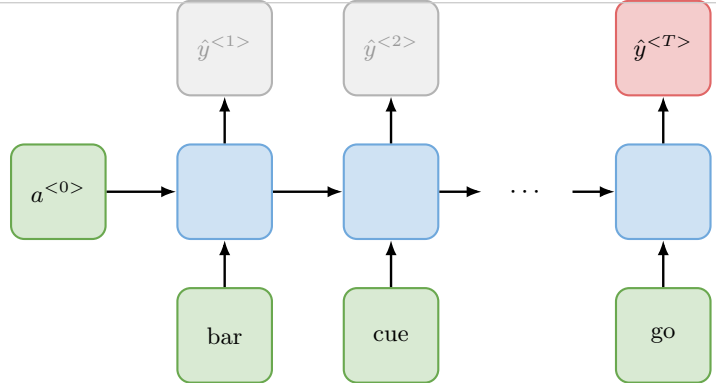
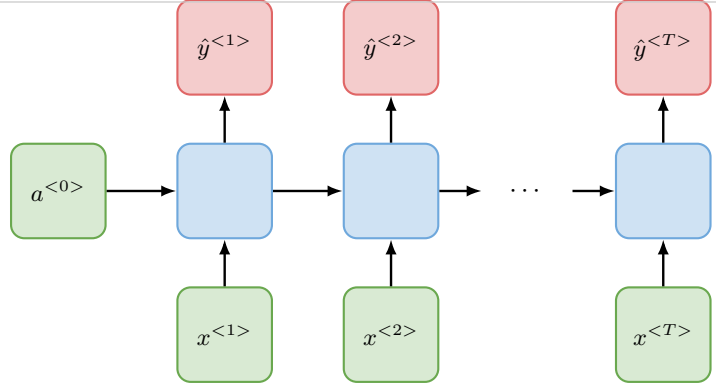


Many-to-many $T_x = T_y$ one step = one trial T = trials in a session		Choice per trial $x^{<t>} = (a_{t-1}, s_{t-1}, r_{t-1}, s_t, \text{cue}_t)$ $\hat{y}^{<t>} = P(a_t)$ over Short / Mid / Long $a^{<t>} = h_t, d = 1-4$ units every step has a label (a_t), so every $\hat{y}^{<t>}$ enters the loss
Many-to-many $T_x = T_y$ one step = 10 ms $T \approx 335$ steps in a trial		Category read-out within the trial $x^{<t>} = u(t) = (\text{bar}(t), \text{rule}(t), \text{go}(t))$ $\hat{y}^{<t>} = z(t)$, category probabilities at step t $a^{<t>} = h(t), N$ rate units same $T_x = T_y$; grey outputs exist but are masked out of the loss until go
Many-to-many $T_x = T_y$ one step = 10 ms same T as row 2		Cursor trajectory within the trial $x^{<t>} = (b(t), \text{dir}, \text{go}(t))$, b from the task network $\hat{y}^{<t>} = (x, y)(t)$, cursor position at step t $a^{<t>} = g(t)$, GRU with M units a measured position at every step: every $\hat{y}^{<t>}$ enters the loss

Notation: $x^{<t>}$ input, $a^{<t>}$ hidden state ($h_t, h(t), g(t)$ in the design document), $\hat{y}^{<t>}$ output, $a^{<0>}$ the learned initial state. $T_x = T_y$ in all three: one output per input step.